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Application Profile

Positioning Statement

I am a nuclear physicist and scientific software developer whose work connects physics analysis, detector systems, and production-scale simulation infrastructure. I am particularly interested in roles that value both scientific judgment and practical software delivery: simulation frameworks, detector operations, computational science, research software engineering, and technical leadership.

Value for Future Roles

- I can bridge domain science and production software: Geant4, [GEMC](#), [CLAS12 Simulations](#) geometry, digitization, and analysis workflows.
- I have experience supporting users through tutorials, examples, technical notes, and operational guidance.
- I understand detector operations and calibration through CLAS12 [Low Threshold Cherenkov Counter](#) work, not only simulation abstractions.
- I have delivered infrastructure for distributed simulation production on HTCondor and Open Science Grid resources.
- I can communicate to multiple audiences: collaborators, students, detector experts, software users, and review committees.

Application Themes

Scientific computing and research software engineering: emphasize [GEMC](#), Geant4, Python/C++, reproducibility, CI, Docker, and user support.

Detector and experimental physics: emphasize [Low Threshold Cherenkov Counter](#), [CLAS12 Simulations](#), [Jefferson Lab Hall-B](#), calibration, detector performance, and nucleon-structure analyses.

Collaboration breadth: where relevant, mention prior work connected to [SPRing-8](#) and dark-sector searches through [Heavy Photon Search](#).

Leadership and user enablement: emphasize tutorials, technical notes, simulation releases, OSG workflows, and cross-collaboration support.

Reusable Short Cover Paragraph

My work at [Jefferson Lab](#) combines nuclear physics, detector systems, and scientific software. I develop and support Geant4/[GEMC](#) simulation workflows, maintain CLAS12 production tools, contribute to detector operation and calibration, and write documentation and tutorials that help users apply complex simulation tools productively. This combination of physics depth, software delivery, and user support is the foundation I would bring to a new role.

Profiles

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